

The Power of Vaccination

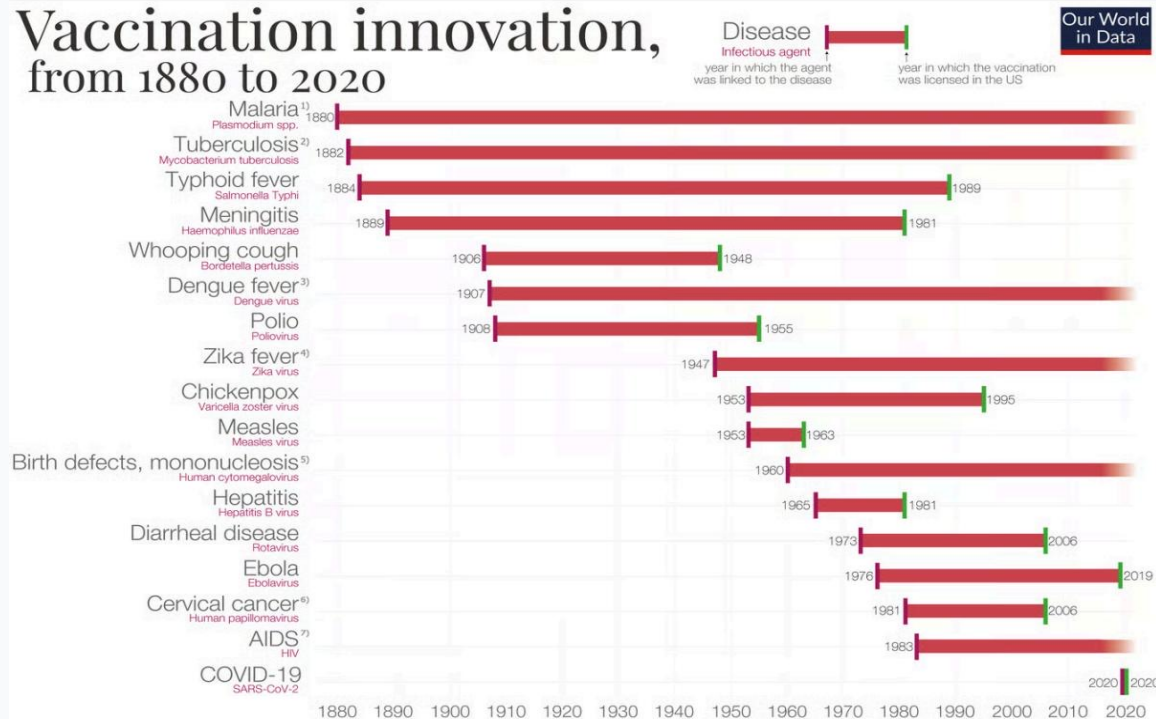
From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of how vaccines have transformed public health (1796–2020)

Source: Our World in Data

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Vaccination innovation, from 1880 to 2020



Pathogen to Licensure Gap

Typhoid Fever

105 Years

(1884–1989)

Measles

10 Years

(1953–1963)

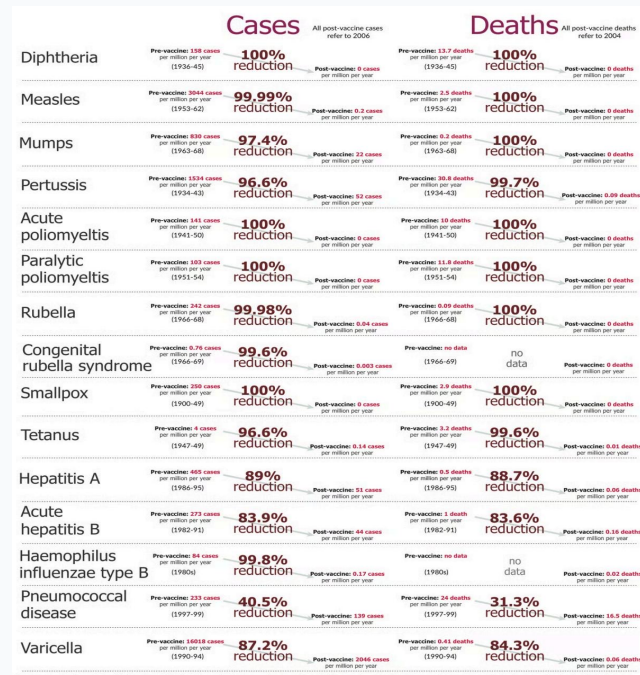
COVID-19

< 1 Year

(2020)

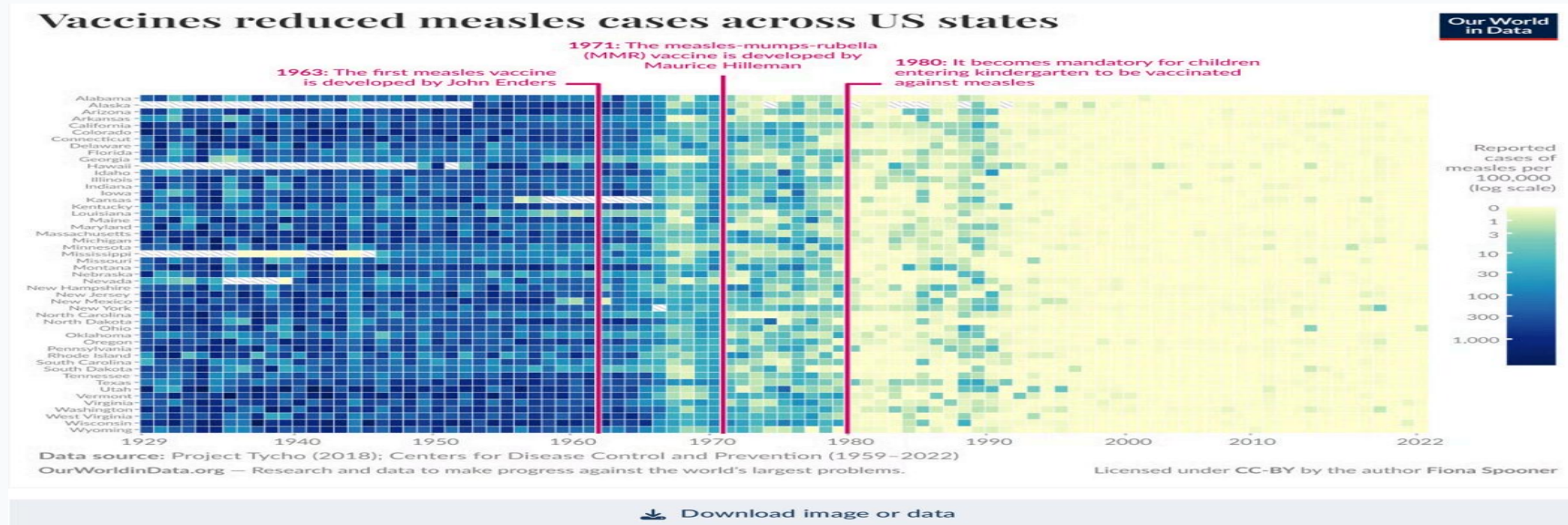
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases	Post-Vaccine Cases	Reduction
Diphtheria	158	0	100%
Smallpox	250	0	100%
Polio (Acute)	141	0	100%
Measles	3,044	0.2	99.99%
Pertussis	1,534	52	96.6%
Hepatitis A	465	51	89.0%
Hepatitis B	273	44	83.9%
Pneumococcal	233	139	40.5%



While many diseases saw near complete elimination, pneumococcal disease and hepatitis control remains incomplete.

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



1963

First measles vaccine developed by John Enders.

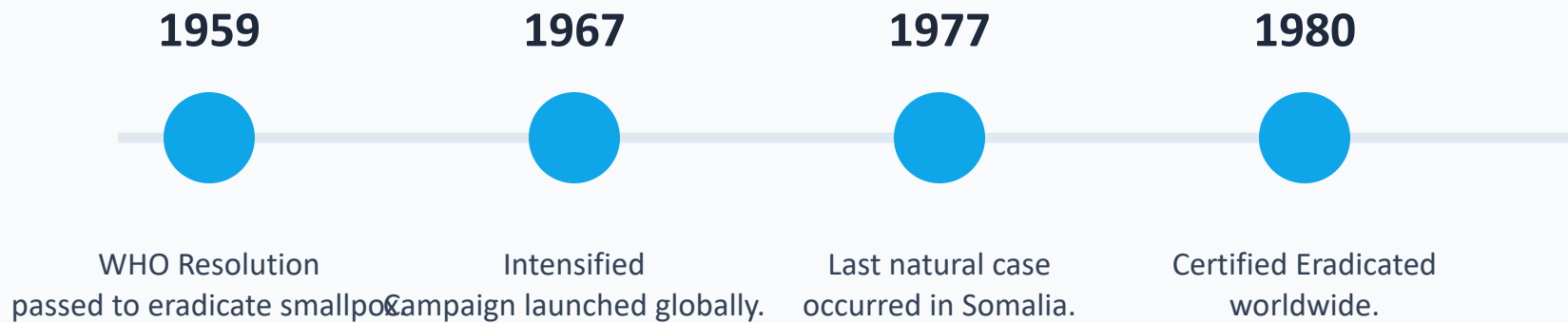
1971

MMR (measles-mumps-rubella) vaccine developed by Maurice Hilleman.

1980

Vaccination became mandatory for kindergarten entry.

Smallpox: The Only Human Disease Eradicated Through Vaccination

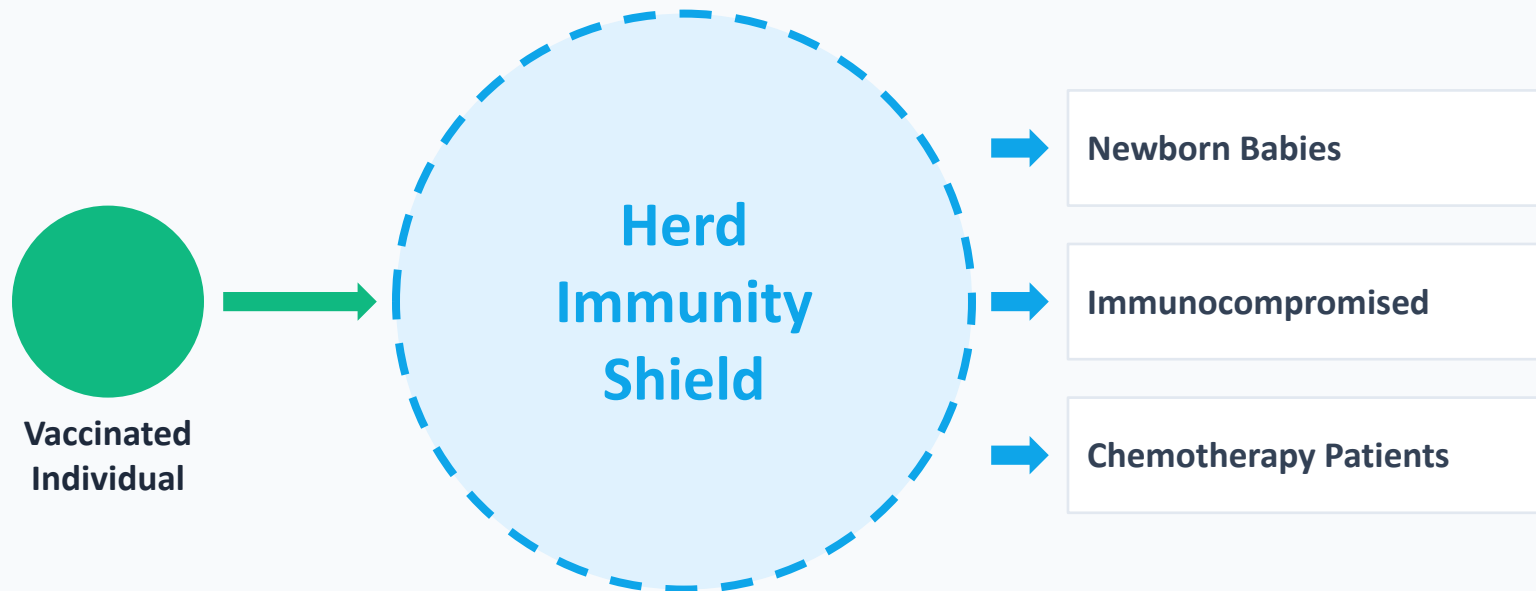


Why Smallpox Was Eradicable

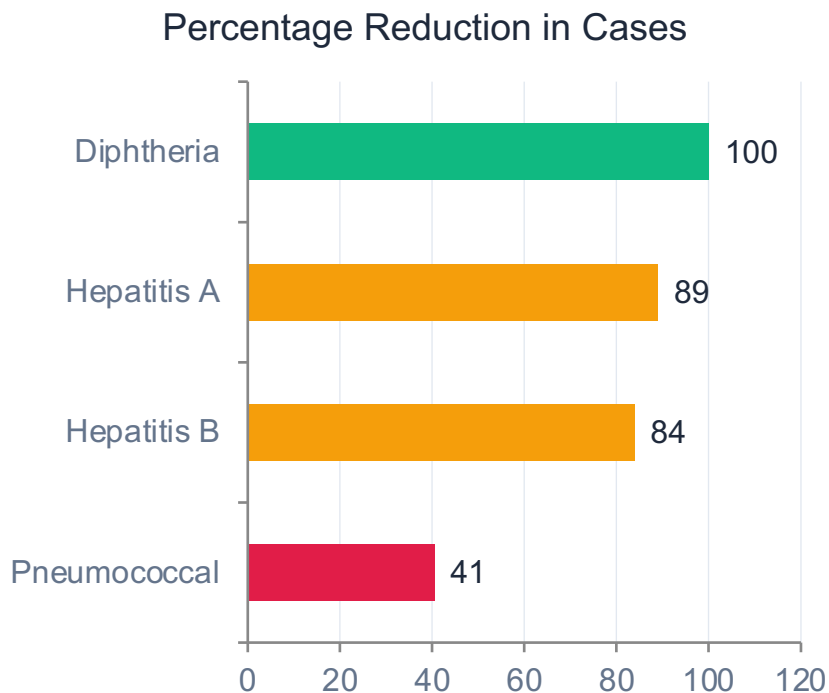
1. Infected only humans
2. Symptoms were clearly distinguishable
3. The vaccine was cheap and effective

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Vaccines protect not only the individual, but also the broader community by halting the spread of pathogens. When vaccination rates are high, an invisible "shield" protects those who cannot be vaccinated.



Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



While many vaccines achieve near 100% elimination, significant challenges remain for several diseases:

Access Gaps

Globally, not everyone has reliable access to life-saving vaccines.

Supply Shortages

Logistical and manufacturing hurdles often put children at risk.

Trust & Misinformation

Poor scientific understanding can reduce overall vaccine uptake.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

100% Elimination

Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the US.

Accelerated Timelines

Development timelines collapsed from over 100 years to under 1 year, driven by mRNA and genomics.

Community Protection

Herd immunity protects entire communities, keeping vulnerable populations safe from outbreaks.

Gaps Persist

Coverage gaps persist for diseases like pneumococcal disease and hepatitis. Global access remains a challenge.

Call to Action: Improving global access, strengthening supply chains, and rebuilding public trust are critical to realizing the full potential of vaccines worldwide.